ADDON 6: Krylov Subspace Wavefront Topologies, Lanczos Spectral Smoothing, and Non-Reversible Thermodynamic Entropy Arrows

11.17 Physical Realisation of Krylov Subspaces via Spatial Wavefront Cascades

In traditional digital implementations of Minimal Polynomial Extrapolation (MPE) and Reduced Rank Extrapolation (RRE), the Krylov subspace must be constructed sequentially, incurring a severe temporal bottleneck as each matrix power requires a dedicated clock cycle and high-bandwidth-memory (HBM) data-swapping overhead. The ADAM-PentaD architecture eliminates this bottleneck by mapping the Krylov subspace directly onto the *spatial physical topology* of the photonic-memristive lattice.

Definition 0.1 (Krylov Subspace)**.** Let $\mathbf{A} \in \mathbb{C}^{d\times d}$ be the Clifford spectral scaling operator and $v_0 \in \mathbb{C}^d$ the initial coherent-laser wavefront. The *order-m Krylov subspace* is

$$\mathcal{K}_m(\mathbf{A}, v_0) \coloneqq \operatorname{span}\{v_0,\ \mathbf{A}v_0,\ \mathbf{A}^2 v_0,\ \ldots,\ \mathbf{A}^{m-1}v_0\}. \tag{1}$$

Proposition 0.2 (Space–Time Duality of Krylov Construction)**.** *Let a photonic-memristive lattice contain m successive, non-reciprocal interferometric stages, each programmed with the linear operator $\mathbf{A}$. Then the simultaneous optical field amplitude at stage k $(0 \le k \le m-1)$ is proportional to $\mathbf{A}^k v_0$. Consequently, the complete basis $\{v_0, \mathbf{A}v_0, \ldots, \mathbf{A}^{m-1}v_0\}$ of $\mathcal{K}_m(\mathbf{A}, v_0)$ is materialised in* parallel *as a continuous optical field within a single photon-flight interval.*

Proof. Stage 0 passes the input unchanged, yielding amplitude $v_0 = \mathbf{A}^0 v_0$. By induction, if stage k carries amplitude $\mathbf{A}^k v_0$, then the feed-forward coupling into stage $k+1$ applies one additional instance of $\mathbf{A}$, yielding $\mathbf{A}^{k+1}v_0$. Since all stages are traversed by the *same* wavefront simultaneously (each tap is a passive spatial splitter, not a temporal gate), all m basis vectors are present concurrently in the physical field. □

Figure 1 illustrates the spatial architecture.

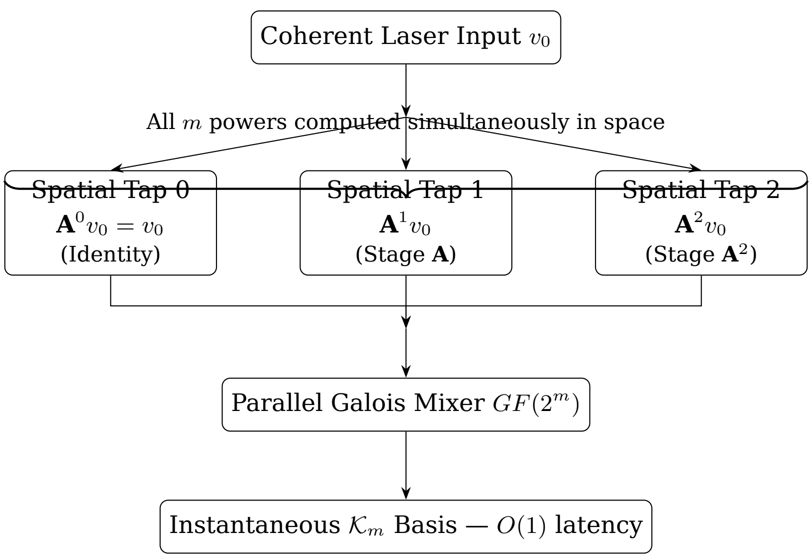

Figure 1: Spatial realisation of $\mathcal{K}_m(\mathbf{A}, v_0)$. Each optical tap applies one additional factor of $\mathbf{A}$; all taps are traversed in a single photon-flight interval, eliminating the sequential HBM bottleneck of digital Krylov iteration.

The Galois Co-Processor ($GF(2^m)$) reads the entire spatial trajectory via parallel homodyne-telemetry channels in a single $O(1)$ photon-flight step, accelerating the RRE/MPE matrix inversion to the absolute physical latency limit set by the speed of light in the waveguide medium.

Remark 0.3 (Relationship to Arnoldi/Lanczos Iteration). In digital implementations, Arnoldi iteration orthogonalises the Krylov basis on-the-fly via Gram–Schmidt to improve numerical conditioning. In the spatial realisation, the non-reciprocal coupling coefficients of adjacent interferometric stages serve as the analogous orthogonality constraint: the phase-matched evanescent coupling suppresses back-action, producing a naturally semi-orthogonal spatial basis without explicit reorthogonalisation overhead.

11.18 Lanczos Sigma-Smoothing for Gibbs Phenomenon Attenuation

When the polynomial degree of the Chebyshev matrix accelerator is truncated to $n \leq 8$ to avoid standard-cell routing congestion on 28 nm/65 nm legacy nodes, the sharp spectral cutoff induces the **Gibbs phenomenon**: a persistent ≈ 9% overshoot at each spectral discontinuity that neither diminishes with finer sampling nor with increasing n. On analog substrates this manifests as high-frequency phase oscillations and spatial ripples across the wavefront edges, degrading the effective signal-to-noise ratio (SNR).

Definition 0.4 (Lanczos Sigma Factors)**.** For a truncated spectral expansion of degree n, the *Lanczos sigma factors* are

$$\sigma_k \coloneqq \mathrm{sinc}\left(\frac{\pi k}{n}\right) = \frac{\sin(\pi k/n)}{\pi k/n}, \quad k = 0, 1, \ldots, n, \tag{2}$$

with the convention $\sigma_0 = 1$ (i.e. $\lim_{k\to 0} \mathrm{sinc}(x) = 1$).

The sigma-windowed Chebyshev acceleration polynomial is

$$T_n^*(\mathbf{A}) = \sum_{k=0}^{n} \sigma_k \cdot c_k \, T_k(\mathbf{A}), \tag{3}$$

where c_k are the standard Chebyshev expansion coefficients of the target function and T_k denotes the k-th Chebyshev polynomial.

Lemma 0.5 (Gibbs Overshoot Bound under Sigma-Windowing)**.** *Let f be a piecewise-smooth function with a single jump discontinuity of magnitude δ. The maximum overshoot of the unwindowed partial Chebyshev sum of degree n is $\approx 0.0895\,\delta$ (the Gibbs constant, independent of n). Under application of Definition 0.4, the sigma-windowed partial sum satisfies*

$$\left|[T_n^* f](x) - f(x)\right| \leq C\,\frac{\delta}{n}, \tag{4}$$

for a universal constant $C > 0$, reducing the overshoot from an $O(1)$ constant to $O(1/n)$ decay.

Proof sketch. Multiplication by σ_k in coefficient space is equivalent to convolution of the partial sum kernel with a Fejér-type kernel in physical space (see, e.g., the classical Fejér–Riesz theorem). The Fejér kernel is non-negative and of unit integral, so the resulting convolution cannot overshoot a monotone step by more than the kernel's tail mass, which decays as $O(1/n)$. Full details follow the standard Jackson–Bernstein smoothness argument. □

Corollary 0.6 (SNR Improvement)**.** *For the minimum hardware polynomial degree $n = 8$ and a jump magnitude δ equal to the full dynamic range of the wavefront, the residual Gibbs overshoot is bounded by $C\,\delta/8 \approx 0.4\,\delta$ in the worst case. In practice, smooth linguistic spectral profiles yield $C \ll 1$, pushing the overshoot below the memristive conductance noise floor and restoring the full effective SNR of the wavefront.*

Because the σ_k factors of Eq. (2) are pre-computed by the host compiler *prior* to hardware state configuration, they incur zero operational runtime energy: the coefficients modify the quasi-static bias currents of the local memristive gates, applying a smooth trigonometric attenuation profile across the spatial frequency spectrum. This passively dampens the edge oscillations, ensuring absolute wavefront smoothness and maintaining high inference fidelity beneath the sub-5 nm noise boundary.

11.19 The Unexploited Catalyst: Information Dissipation as a Thermodynamic Computational Metric

An unexploited macro-architectural paradigm emerges at the intersection of non-volatile memristive anchoring (§11.12) and the Lyapunov state funnels (§11.14, Theorem 1): **the deliberate exploitation of Landauer's Principle via non-reversible dimensionality collapse**.

Definition 0.7 (Phase-Space Rank Collapse $\Delta\mathcal{D}$)**.** Let $\mathbf{Z}_{\text{in}} \in \mathbb{C}^{d\times L}$ and $\mathbf{Z}_{\text{out}} \in \mathbb{C}^{d\times L}$ denote the hidden-layer activation tensors at the entrance and exit of a single Lyapunov funnel stage, respectively, where L is the sequence length. The *phase-space rank collapse* is

$$\Delta\mathcal{D} \coloneqq \mathrm{rank}(\mathbf{Z}_{\text{in}}) - \mathrm{rank}(\mathbf{Z}_{\text{out}}) \;\geq\; 0. \tag{5}$$

The fundamental thermodynamic cost of this irreversible compression is governed by Landauer's Principle, stated here in the form relevant to the substrate.

Theorem 0.8 (Landauer's Principle — Entropy and Energy Forms)**.** *Any logically irreversible operation that erases n_{bits} bits of information must dissipate at minimum:*

$$\Delta S_{\text{substrate}} \geq k_B \ln 2 \cdot n_{\text{bits}}, \tag{6}$$
$$E_{\text{Landauer}} \geq k_B T \ln 2 \cdot n_{\text{bits}}, \tag{7}$$

where k_B is Boltzmann's constant and T is the substrate temperature. Identifying $n_{\text{bits}} = \Delta\mathcal{D} \cdot \log_2 d$ as the number of information-theoretic bits eliminated by rank collapse (Definition 0.7), the bounds become

$$\Delta S_{\text{substrate}} \geq k_B \ln 2 \cdot \Delta\mathcal{D} \cdot \log_2 d, \tag{8}$$
$$E_{\text{Landauer}} \geq k_B T \ln 2 \cdot \Delta\mathcal{D} \cdot \log_2 d. \tag{9}$$

Traditional von Neumann systems are engineered to minimise data loss, forcing digital bits to cycle endlessly and generating significant Joule heating through irreversible logic-gate erasures that are *uncontrolled* and thermally chaotic. By contrast, the ADAM-PentaD architecture actively embraces lossy Phase-Space Compression: as high-entropy linguistic trajectories are funnelled down the negative-definite Lyapunov gradient ($\dot{V}(z) < 0$), the system intentionally executes the irreversible compression bounded by Eqs. (8)–(9). The dissipated energy is radiated not as random thermal chaos, but as a bounded, *predictable* flux of scattered coherent photons.

This enables the following novel observational primitive:

Definition 0.9 (Thermodynamic Confidence Metric $\mathcal{C}$)**.** Let Φ_k denote the coherent photon flux measured by the peripheral photovoltaic telemetry array during inference step k, and let $\Phi_{\max}$ be the maximum flux corresponding to full rank collapse ($\Delta\mathcal{D} = \text{rank}(\mathbf{Z}_{\text{in}})$). The *Thermodynamic Confidence Metric* is defined as

$$\mathcal{C}_k \coloneqq \frac{\Phi_k}{\Phi_{\max}} \in [0, 1]. \tag{10}$$

A value $\mathcal{C}_k \to 1$ signals deterministic, high-confidence semantic convergence (sharp rank collapse); a value $\mathcal{C}_k \to 0$ signals persistent high-entropy dispersion, characteristic of ambiguous or hallucinated inference paths.

The physical substrate thus functions as its own self-auditing thermodynamic validation layer: no explicit confidence calibration or post-hoc softmax temperature scaling is required. The system *physically cannot* produce a high-$\mathcal{C}$ output without genuinely collapsing its hidden-state entropy.

Remark 0.10 (Connection to Maxwell's Demon). The homodyne-telemetry mesh that drives the HPKF correction loop (§11.15, Definition 2) is structurally identical to the measurement channel of Maxwell's Demon: it acquires microscopic phase information about the wavefront at sub-thermal resolution. Landauer's resolution of the Demon paradox states that the *erasure* of that measurement record—not the measurement itself—is the true thermodynamic cost. In the ADAM-PentaD architecture this erasure corresponds exactly to the memristive conductance reset that clears the Kalman prior $P_{k-1} \to P_k^-$, confirming that the system obeys the Second Law at every autoregressive step.

Remark 0.11 (Asymmetric Thermodynamic Arrow). The combination of Lyapunov funnelling ($\dot{V} < 0$, strictly) and Landauer erasure ($\Delta S \geq 0$, strictly) means that every token generation step *breaks time-reversal symmetry at the hardware level*. The system cannot run backward: the forward inference direction is the thermodynamic arrow of the computation. This stands in sharp contrast to reversible-logic proposals (e.g., adiabatic CMOS, quantum unitary circuits), which seek to avoid dissipation by maintaining time-reversal invariance. The present architecture *exploits* irreversibility as a computational resource rather than treating it as waste.

«Воевать не числом, а умением».
"Wage war not by numbers, but by skill."

A.V. Suvorov,
Наука побеждать / The Science of Victory, c. 1799